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DRAFT — Universal anchor § . . : Conflict-free multi-track work-division + exactly-once claim registry + capability-aware deadlock-proof device-lock (REM- + REM- + REM-)

Revision: 1 **Last modified:** 2026-07-02T23:45:00Z **Status:** READY-TO-LAND draft (NOT yet landed). Conductor lands via §11.4.26 (fetch-first → reconcile final number → paste into the 5 carriers → propagation gate + paired mutation → push all upstreams ff-only §11.4.113). This file is authored by Stream A per §11.4.101 (safe reversible artifact instead of a high-blast-radius half-edit of the 5 live carriers under an active rate-limit). **Classification: universal (§11.4.17)** — a project-neutral multi-track parallel-development safety mechanism reusable by ANY project running N work-streams over M shared resources; the consuming project supplies its concrete claim-store, device pool, and resource-metric sources per §11.4.35. Nothing ATMOSphere-specific lands in the canonical root (REM-03). **Proposed number:** §11.4.176 (both §11.4.175 and §11.4.176 verified FREE 2026-07-02; §11.4.175 is P0_PLAN-reserved — conductor reconciles at §11.4.26 fetch-first time).

Canonical Constitution.md section (paste-ready)

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§ . . — Conflict-free multi-track work-division + exactly-once claim registry + capability-aware deadlock-proof device-lock (User mandate, - -)

Forensic anchor — verbatim user mandate (2026-07-02, REM-17): "MUST prevent multiple Tracks working the same workable item OR the same group of workable items resembling one consistent logical group of functionalities. Division of work assigned to Tracks MUST be conflict-free: no conflicts, misunderstandings, race conditions, data corruptions, codebase loss. Risk-free + safe; results ALWAYS produced; no false results, no bluff. EXTEND the constitution submodule with all these critical details for full universal reusability by all projects." (REM-02): "there MUST exist proper synchronization mechanism for flashing and use of device ... lock ... release ... available test devices pool and information about held locks ... dead-lock-proof ... a track locks one OR more devices selected by required capability."

When a project runs multiple concurrent work-STREAM tracks (§11.4.167 feature work-stream lifecycle; §11.4.58/§11.4.103 parallel streams) that share (a) a common pool of workable items and (b) a scarce pool of exclusive hardware/ resources, TWO cooperating arbitration layers are MANDATORY. Both are conflict-free BY CONSTRUCTION, not by convention.

(A) Work-item layer — exactly-once claim registry (REM-17). Every workable item, OR every logical GROUP of items that share files / subsystem / feature / schema (a hidden-coupling cohort), is assigned to AT MOST ONE track at a time. (1) **Exactly-once** — a lock-arbitrated, append-only claim registry (extending §11.4.147 no-work-loss registry + §11.4.116 real-time sync substrate) records `claim(item|group, track, ts, ttl)`; a track MUST claim-before-touch; a second claim on an already-claimed item is DENIED (no silent overlap). (2) **Logical-group cohesion** — items that share a file-scope / subsystem / feature / schema are one group and are claimed WHOLE by one track (prevents the hidden-coupling race two tracks editing the same seam would create — §11.4.58 L3 disjoint-file-scope is the enforcement). (3) **Conflict-free by construction** — disjoint

file-scope partition (§11.4.58 L3) + per-track worktree/CoW isolation (§11.4.167) means two tracks physically cannot touch the same bytes. (4) **No codebase loss** — §9.2 hardlinked backups, §11.4.113 ff-only merge-onto-latest-main (NEVER force-push), §11.4.84 working-tree quiescence, §11.4.41 merge-first. (5) **Anti-bluff** — the assignment + the merge are verifiable (registry event stream + captured merge evidence); results are ALWAYS produced (a claimed-but-abandoned item's claim TTL-expires and returns to the pool, §11.4.147).

(B) Hardware layer — capability-aware deadlock-proof device-lock (REM-02). When N tracks outnumber M exclusive test devices, exactly ONE track owns each device at a time (§11.4.119 single-resource-owner). A track leases ONE OR MORE devices selected by required CAPABILITY (not merely identity) — the pool records per-device capabilities; leases are capability-matched; multi-device leases are supported. The arbiter is DEADLOCK-FREE by construction — all four Coffman conditions broken: hold-and-wait ELIMINATED (multi-device acquire is ALL-OR-NOTHING inside one critical section — never a partial hold while waiting); circular-wait ELIMINATED (acquire is NON-BLOCKING — wins the whole set at once or returns EBUSY immediately, never waits on another track's device); no-preemption RELAXED (lease TTL + heartbeat + reap — a crashed holder's lease auto-expires, §11.4.147, so a dead track never holds the pool hostage); the only lock ever held is a short bounded flock over the atomic check-and-claim critical section, released before any device work. Registry = append-only JSONL event stream + atomically-rewritten snapshot (§11.4.116). **Reference implementation (this project):** `scripts/multitrack/multitrack_device_lock.sh` + its anti-bluff stress/chaos/paired-mutation proof `scripts/multitrack/test_multitrack_device_lock.sh` (8/8 PASS, deterministic ×2, mutation CAUGHT: guard-removed arbiter double-leases 12/round vs 1/round for the real arbiter).

(C) Universal decoupling + evidence-based resource-tuning (REM-03). Both layers are universal, decoupled, project-not-aware, reusable (§11.4.17/§11.4.28/§11.4.35) — the consuming project injects its claim-store, device pool, and metric sources at runtime; nothing project-specific lands in the canonical root. Resource-tuning decisions (parallel-stream count, per-stream memory budget §12.6, device-lease breadth, build-queue depth §11.4.167) MUST be EVIDENCE-BASED — driven by captured metrics (§11.4.24 build-resource stats / §11.4.5 captured evidence), never guessed (§11.4.6). "Run more tracks / bigger `-j`" without captured headroom evidence is a §11.4.6 + §12.6 violation.

Distinct-but-composing: (A) arbitrates WORKABLE-ITEM/task assignment; (B) arbitrates HARDWARE; they are the two multi-track safety layers and compose (an alias/account resource dimension, REM-18, is a third that reuses the same registry pattern).

Composes §11.4.6 / §11.4.17 / §11.4.24 / §11.4.28 / §11.4.35 / §11.4.41 / §11.4.58 / §11.4.84 / §11.4.85 / §11.4.103 / §11.4.113 / §11.4.116 / §11.4.119 / §11.4.147 / §11.4.167 / §12.6 / §9.2. Propagation gate `CM-COVENANT-114-176-PROPAGATION` (literal `11.4.176` across the consumer fleet) + recommended gates `CM-WORK-DIVISION-EXCLUSIVE-CLAIM` (no item/group claimed by >1 track) + `CM-NO-CROSS-TRACK-SCOPE-OVERLAP` (per-track file-scope manifests are pairwise disjoint) + `CM-LOGICAL-GROUP-COHESION` (shared-seam items are one group, claimed whole) + `CM-DEVICE-LOCK-DEADLOCK-FREE` (device-lock arbiter ships an all-or-nothing + non-blocking + TTL-reap stress/chaos proof with a paired §1.1 mutation) + paired §1.1 meta-test mutations (gate-code = separate work item). No escape hatch — no `--allow-cross-track-overlap`, `--skip-claim-registry`, `--split-logical-group`, `--blocking-device-acquire`, `--allow-partial-device-hold`, `--tune-resources-by-guess` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.176. Non-compliance is a release blocker regardless of context.

Paste-ready compact mirror line (for CLAUDE.md / AGENTS.md / QWEN.md / GEMINI.md anchor-list sections)

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- §11.4.176 — Conflict-free multi-track work-division + exactly-once claim registry + capability-aware deadlock-proof device-lock (User mandate, 2026-07-02).** Two mandatory multi-track arbitration layers, conflict-free by construction: (A) exactly-once work-item/logical-group claim registry (extends §11.4.147/§11.4.116; disjoint file-scope §11.4.58 L3 + worktree/CoW §11.4.167; no codebase loss §9.2/§11.4.113/§11.4.84/§11.4.41); (B) capability-aware deadlock-proof device-lock (§11.4.119 single-owner; all-or-nothing + non-blocking + TTL-reap breaks all 4 Coffman conditions; append-only JSONL + atomic snapshot §11.4.116); (C) universal/decoupled (§11.4.17/§11.4.28/§11.4.35) + evidence-based resource-tuning (§11.4.24/§11.4.6, never guessed). [full → CLAUDE_ANCHORS_FULL.md + Constitution.md] · refs: §11.4.6 §11.4.17 §11.4.24 §11.4.28 §11.4.35 §11.4.41 §11.4.58 §11.4.84 §11.4.85 §11.4.103 §11.4.113 §11.4.116 §11.4.119 §11.4.147 §11.4.167 §12.6 §9.2 §11.4.176
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§ . . conductor landing checklist (do NOT skip a step)

1. **Fetch-first** — inside `constitution/` : `git fetch --all --prune` every remote; `git pull --ff-only` . Re-confirm §11.4.176 is STILL free (P0_PLAN's §11.4.175 or another anchor may have landed); if taken, bump to the next free integer and update every occurrence in this draft + the mirror line before pasting.
2. **Paste** the §11.4.176 section into `constitution/Constitution.md` after the current last anchor; append the compact mirror line to the anchor-list section of `constitution/CLAUDE.md` , `constitution/AGENTS.md` , `constitution/QWEN.md` , `constitution/GEMINI.md` (5 carriers, §11.4.35/§11.4.157). Add the full text to `constitution/CLAUDE_ANCHORS_FULL.md` if that expansion file is in use.
3. **Bump** each carrier's revision-header `Revision:` + `Last modified:` (§11.4.44); regenerate HTML/PDF/DOCX exports (§11.4.65/§11.4.73 pipeline).

4. **Wire** the propagation gate `CM-COVENANT-114-176-PROPAGATION` + the 4 recommended gates + their paired §1.1 mutations (gate-code = a separate tracked work item; may land in a follow-up commit).
5. **Validate** — no conflict markers; `meta_test_inheritance.sh` (or equivalent); re-run pre-build.
6. **Commit + push to ALL upstreams** (§2.1) ff-only (§11.4.113 — NEVER force-push); bump the consuming project's submodule pointer in the same commit (§11.4.26 step 7).
7. **Delete this draft** once landed (its content now lives in the live carriers).